

L' Hospital's Rule:

Lee Calculus

Undetermined quantities

$$\frac{0}{0}, \frac{\infty}{\infty}, 0(\infty), (\infty - \infty), (0)^0, (0)^\infty, (1)^\infty$$

Let $f(x)$ & $g(x)$ be differentiable

functions. If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$

$$\text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided that $g'(x) \neq 0$ near $x = a$.

$$\text{Ex. } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{(1)} = \frac{0}{1} = 0$$

$$\text{Ex. } \lim_{x \rightarrow \infty} \frac{2x^2 + 6x + 1}{7x^2 + x - 3} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{4x + 6}{14x + 1} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{4}{14} = \frac{4}{14} = \left[\frac{2}{7} \right]$$

$$\text{Ex. } \lim_{x \rightarrow 1} \frac{\ln x}{x-1} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{1} = \frac{1}{1} = [1]$$

$$e^{\infty} = \infty, e^{-\infty} = 0$$

$$e^0 = 1$$

$$\ln \infty = \infty$$

$$\ln 0 = -\infty$$

$$\ln 1 = 0$$

Remark:

$$\lim_{x \rightarrow 0} \frac{\ln x}{x-1} = \frac{\ln 0}{-1} = \frac{-\infty}{-1} = \frac{\infty}{1} = \infty$$

You can not apply
L'Hospital

"it is not $\frac{0}{0}$ or $\frac{\infty}{\infty}$ "

$$\text{Ex. } \lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \frac{e^\infty}{\infty} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{2x} = \frac{\infty}{2(\infty)} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{2} = \frac{\infty}{2} = \infty$$

$$\text{Ex. } \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} = \frac{0 - 0}{0} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} = \frac{1 - 1}{0} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sec x \sec x \tan x}{6x} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{2(2) \sec x \sec x \tan x \tan x + 2 \sec^2 x \sec^2 x}{6}$$

$$= \frac{0 + 2}{6} = \left[\frac{1}{3} \right]$$

$$\text{Ex. } \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} = \frac{\ln \infty}{\sqrt[3]{\infty}} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{3} x^{-2/3}} = 3 \lim_{x \rightarrow \infty} \frac{x^{2/3}}{x}$$

$$= 3 \lim_{x \rightarrow \infty} \frac{1}{\sqrt[3]{x}} = 3 \cdot \frac{1}{\infty} = 0$$

Ex. $\lim_{x \rightarrow 0^+} x \ln x = 0 \ln 0 = 0 \quad (-\infty)$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\left(\frac{1}{x}\right)} = \frac{\ln 0}{\left(\frac{1}{0}\right)} = \frac{-\infty}{\infty}$$

$$= \lim_{x \rightarrow 0^+} \frac{1/x}{\left(-\frac{1}{x^2}\right)} = - \lim_{x \rightarrow 0^+} \frac{x^2}{x}$$

$$= - \lim_{x \rightarrow 0^+} (x) = -(0) = 0$$

Ex. $\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x) = \frac{1}{\cos \frac{\pi}{2}} - \frac{\sin \frac{\pi}{2}}{\cos \frac{\pi}{2}}$

$$= \frac{1}{0} - \frac{1}{0}$$

$$= \infty - \infty$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{1}{\cos x} - \frac{\sin x}{\cos x} = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{1 - \sin x}{\cos x}$$

$$= \frac{0}{0}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{-\cos x}{-\sin x} = \frac{0}{1} = \boxed{0}$$

Ex. $\lim_{x \rightarrow 0^+} x^x = (0)^0$

let $y = x^x$

$\ln y = x \ln x$

$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x = 0(-\infty)$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{(1/x)} = \frac{1/x}{(-1/x^2)}$$

$$= \lim_{x \rightarrow 0^+} (-x) = 0$$

then $\lim_{x \rightarrow 0^+} y = e^0 = 1$

$$\Rightarrow \lim_{x \rightarrow 0^+} x^x = \boxed{1}$$

Ex. $\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x} = (1+0)^\infty = (1)^\infty$

let $y = (1 + \sin 4x)^{\cot x}$

$\ln y = \cot x \ln(1 + \sin 4x) = \frac{\ln(1 + \sin 4x)}{\tan x}$

$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 4x)}{\tan x} = \frac{0}{0}$

$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{1 + \sin 4x} \cdot 4 \cos 4x}{\sec^2 x}$

$= \lim_{x \rightarrow 0^+} \frac{4 \cos 4x}{\sec^2 x (1 + \sin 4x)} = \frac{4}{(1)(1)} = 4$

then $\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x} = \boxed{e^4}$

Example: Find the following limits

(a) $\lim_{x \rightarrow 0} \frac{\tanh 7x}{x}$

(b) $\lim_{x \rightarrow \infty} x e^{-x}$

(c) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{3x}\right)^{4x}$

Solution:

(a) $\lim_{x \rightarrow 0} \frac{\tanh 7x}{x} = \frac{0}{0} = \lim_{x \rightarrow 0} \frac{7 \operatorname{sech}^2 7x}{1} = \frac{7(1)}{1} = \boxed{7}$

(b) $\lim_{x \rightarrow \infty} x e^{-x} = 0(\infty) = \lim_{x \rightarrow \infty} \frac{x}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{\infty} = \boxed{0}$

(c) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{3x}\right)^{4x} = (1)^\infty$

let $y = \left(1 + \frac{1}{3x}\right)^{4x} \Rightarrow \ln y = 4x \ln \left(1 + \frac{1}{3x}\right)$

$\ln y = \frac{\ln \left(1 + \frac{1}{3x}\right)}{1/4x} \Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{3x}\right)}{1/4x} = \frac{0}{0}$

$= \lim_{x \rightarrow \infty} \frac{\left(\frac{-1}{3x^2}\right)}{\left(1 + \frac{1}{3x}\right)\left(\frac{-1}{4x^2}\right)} = \lim_{x \rightarrow \infty} \frac{4/3}{1 + \frac{1}{3x}} = 4/3$

then $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{3x}\right)^{4x} = e^{4/3}$

Page

61